Mst Afsana Mimi

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Summary

Statistics master's graduate with experience preparing multi-source datasets, building analytical workflows, and translating quantitative findings into clear recommendations. Proficient in SQL, Python, and R, with strengths in data quality, trend analysis, statistical modeling, and reproducible reporting. Interested in applying rigorous analytics to operational and financial-services decisions.

Education

University of Massachusetts Amherst
Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

University of Dhaka
Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

University of Dhaka
Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Technical Skills

Programming & Querying

- SQL (joins, aggregations, window functions), Python (Pandas, NumPy, Scikit-learn), R

Data Preparation & Analytics

- Generative AI concepts: LLM workflows, retrieval-augmented generation (RAG), prompt design, evaluation, and safety guardrails
- ETL/ELT workflows, data preprocessing, integration, and dataset construction
- Data validation, consistency checks, trend analysis, and quantitative reporting
- Regression, hypothesis testing, statistical modeling, and model evaluation

Reporting & Platforms

- Tableau, data visualization, Git/GitHub
- AWS (S3, EC2) – familiarity

Experience

Teaching Assistant (Data & Statistics)
UMass Amherst Sep 2023 – May 2026

- Supported student projects involving R/Python-based data analysis.
- Explained statistical results clearly and helped students develop evidence-based conclusions.

Research Assistant (NLP & Data Analysis)
University of Dhaka Jul 2021 – Jun 2022

- Integrated and transformed large-scale text data through reproducible ingestion and preprocessing workflows.
- Developed analyses and machine-learning models in Python to identify patterns and evaluate performance.
- Applied validation and consistency checks to improve data quality and the reliability of reported findings.

Research Assistant (Socio-economic Data Analysis)
University of Dhaka Jul 2018 – Jun 2019

- Cleaned, combined, and structured large-scale survey datasets for reproducible statistical analysis.
- Analyzed socio-economic trends and converted model results into findings for policy-focused research.
- Maintained version-controlled analysis workflows and documented data-processing steps for collaboration.

Projects

Can Agentic AI Eliminate the Need for a Dedicated Relational Foundation Model?

- Developing an agentic framework that interprets relational schemas, constructs evidence across linked tables, and performs predictive reasoning over structured data.
- Evaluating relational workflows and designing a benchmark for realistic database-analysis tasks.

Gaussian Process Modeling with Fairness Constraints

- Developing Gaussian Process models with fairness constraints across demographic groups.
- Designing evaluation strategies to assess trade-offs between predictive accuracy and bias.

High-Dimensional Feature Selection (MCAR)

- Developed a feature-selection method for high-dimensional censored data and evaluated it through simulations and biomedical data.
- Compared performance with established approaches; results were published in *Statistics in Medicine*.

Awards and Scholarship

- Conference Awards for Scientists (CASc) by International Society for Clinical Biostatistics (ISCB).
- National Science and Technology (NST) Fellowship Award by Education Board of Bangladesh.
- Dhaka University Merit Scholarship for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra